

DLE602 Assessment 1: Twitter Sentiment Analysis Report

Student: Nomayer Hossain

ID: A00176827

Subject: DLE602 Deep Learning

Google Drive:

https://drive.google.com/drive/folders/1qXpoUzllfLgqa6vtSnmJTT5h_bRUN79o?usp=sharing

This report presents a Twitter sentiment analysis system built around a Bigram N Gram probabilistic language model, following the classification rule specified in the assessment brief: a tweet is labelled positive if at least one quarter of its tokens are positive sentiment words, negative if at least one quarter are negative, and neutral otherwise. Zhao, Gui and Zhang (2018) evaluated their models on five named Twitter datasets. Two of these, the Stanford Twitter Sentiment Gold set and the SemEval 2014 Task 9 corpus, are no longer accessible online, since the former was taken offline in 2024 and no working public source could be found for the latter. To keep the assessment feasible while staying close to the paper's original data source, two samples were instead drawn from the Sentiment140 corpus (Go, Bhayani and Huang, 2009), the training set behind the Stanford Twitter Sentiment Test dataset used by Zhao et al. Dataset 1 is a balanced random sample of 800 general tweets. Dataset 2 is a balanced sample of 800 tweets filtered for product or service keywords, representing a topic focused population comparable to the paper's domain specific sets.

Two issues were identified and corrected during development. First, the preprocessing pipeline removed English stopwords before generating bigrams, and negation words such as "not", "never" and "no" are themselves classified as stopwords by NLTK. Negation words were therefore stripped before the bigram negation handler ever saw them, so phrases such as not happy were never correctly inverted to negative. The fix retains negation trigger words through preprocessing so the bigram model can apply polarity inversion as intended. Second, the sentiment lexicon initially used a small, hand-picked set of around 420 words. Since Zhao et al. cite the AFINN word list (Nielsen, 2011) as their sentiment feature source, the lexicon was replaced with the full AFINN 165 list of over 3,300 scored words, extended with a short list of Twitter specific slang absent from the formal lexicon.

After both fixes, overall accuracy reached 26.0 percent for Dataset 1 and 20.2 percent for Dataset 2. Positive precision was 72.6 percent and 82.0 percent respectively, and negative precision was 90.7 percent and 85.7 percent, showing the model is usually correct when it does commit to a label. Recall stayed low, between 12 and 33 percent across both classes, because most tweets lacked enough explicit

sentiment words to reach the 25 percent threshold and defaulted to neutral. Since Sentiment140 contains no true neutral tweets, every neutral prediction is automatically incorrect, capping accuracy regardless of lexicon quality.

The two datasets behaved consistently rather than differently. Both show the same pattern of high precision paired with low recall and a majority neutral prediction rate, 67 percent for Dataset 1 and 76 percent for Dataset 2. Dataset 2 performed slightly worse overall, likely because topic focused tweets about products and services contain more descriptive, technical language and fewer explicit emotion words than general tweets. This suggests the accuracy ceiling is a structural property of a fixed threshold, single token lexicon approach applied to short, sparsely emotional text, rather than a weakness specific to either dataset. Overall, the model demonstrates consistent classification behaviour across both sources, even though its accuracy remains modest given the strictness of the specified threshold rule.

References

- Go, A., Bhayani, R. and Huang, L. (2009). Twitter sentiment classification using distant supervision. Stanford University, CS224N Technical Report.
- Nielsen, F. Å. (2011). A new ANEW: Evaluation of a word list for sentiment analysis in microblogs. Proceedings of the ESWC2011 Workshop on Making Sense of Microposts.
- Zhao, J., Gui, X. and Zhang, X. (2018). Deep convolution neural networks for Twitter sentiment analysis. IEEE Access, 6, 23253 to 23260.